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Customizable and Foldable Cane

Our group proposes the development of a customizable ergonomic cane designed to enhance mobility and comfort for individuals requiring assistance with walking. This design will have 3 moving parts with 2 hinges and a ball and socket joint that rotates. This biomedical device aims to address the limitations of traditional canes by incorporating user-friendly features and advanced functionality.

Shaft

Our design focuses on three key improvements. The first enhancement is a foldable shaft with two hinges, effectively breaking the cane into three parts. This design enhances portability but also offers storage benefits. We are looking into adding a height adjustment on the shaft if the default size isn't suitable but want to do an alternative to the spring-loaded mechanism traditional canes use. Right now, we are looking at a tightening channel for our height adjustment.

Base

The second improvement involves the incorporation of a ball and socket mechanism at the base of the cane. This feature enables micro-adjustments, particularly beneficial when navigating uneven terrain and providing stability [1]. By providing stability and adaptability, our cane ensures a smoother and safer walking experience for users across various environments. To address the varying needs of users, our cane offers a quad shape. Researchers from the Journal and Prosthetics and Orthotics claim that quad canes are much more effective in improving postural steadiness compared to straight canes [2]. These findings contribute to adding the quad cane feature to our design.



Handle



We have redesigned the handle of the cane to optimize comfort and functionality. The handle now

features an oval shape, maintaining a flat top surface that aids in standing up from a seated position. An article from Mayo Clinic states that the handle of a walking cane is most comfortable for the user when your elbow is at a 15-20 degree angle when holding it [3]. Simultaneously, the ergonomic sides of the oval offer a more comfortable and secure grip during walking, reducing strain on the user's hand and wrist. Although we are not opposed to having customizable options such as a Fritz handle. In a research paper written for the 8th International Conference on Industrial, Mechanical, Electrical and Chemical Engineering (ICIMECE 2023) about the measurement of forearm muscle activity and elbow angle of the elderly in using different types of cane handles, they found that the Fritz handle with the posterior position and the initial angle has a minimal load distribution from the forearm to the wrist so that it can be a selection suggestion to the elderly [4].



Fritz Handle

References

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